

SQL Sales Data Analysis using Subqueries, CTEs, and Window Functions. (Detailed Report Of Week 3)

Objective

Analyze Superstore sales data using SQL concepts such as Subqueries, Common Table Expressions (CTEs), and Window Functions to answer business-related questions and generate meaningful insights.

Dataset

- Superstore Dataset
- Table: superstore_raw

Tasks Performed

1. Data Preparation-

- Created superstore_raw table.
- Loaded Superstore dataset into the database.
- Created dimensional tables:
 - customers
 - products
 - orders
- Inserted data using SELECT DISTINCT.

2. Subqueries are performed such as -

- Customers with sales above average sales.
- Highest-value order for each customer.

3. Common Table Expressions (CTEs) such as-

- Calculated total sales per customer.
- Calculated total profit per customer.

4. Window Functions like-

- ROW_NUMBER() for sequential numbering.
- RANK() for customer ranking based on sales.
- DENSE_RANK() for ranking without gaps.

5. JOIN + CTE + Window Functions

- Combined customer information with aggregated sales data.
- Ranked customers according to total sales contribution.

6. Business Queries

- Top 10 customers by sales.
- Bottom 10 customers by sales.
- Customers with exactly one order.
- Customers whose total sales are above average.

- Highest order value per customer.

Key Insights

- A small group of customers contributes the highest share of revenue.
- Several customers have only one purchase, indicating potential retention opportunities.
- Above-average customers can be targeted for loyalty programs.
- Ranking analysis helps identify high-value customers for business decisions.

Technologies Used

- Python (Pandas, SQLite3)
- SQL — Subqueries, CTEs, Window Functions, JOINS
- Jupyter Notebook
- Dataset: Sample Superstore CSV

Outcome-

Successfully applied Subqueries, CTEs, Window Functions, and Joins to perform sales analysis and answer business questions using the Superstore dataset.